Mathematical Economics.

Simultaneous differential equations:

A differential equation is an equation which contains one or more ~~abcm~~ terms and the derrivatives of one variable (dependent variable) with respect to the other variable (independent variable).

$$\frac{dy}{dx} = f(x).$$

Here "x" is an independent variable "y" is an dependent variable.

Example:- $\frac{dy}{dx} = 5x$

Simulltaious difference equation:-

A simulltainious difference equation is one of the mathematical equat for an indefinit function of more than one variables, that relates the values of the function.

Importance:

Difference equation play an important role and function in engineering, Physics, economics and other disciplines.

Differential equation.

Formula:-

$$A_1 \dot{Y} + A_2 Y = B.$$

For $r_1, r_2 = (A_1 r_1 + A_2) = 0.$

eigenvector $= (A_1 r_i + A_2) C_i = 0.$

$$Y_c = K_1 C_1 e^{r_1 t} + K_2 C_2 e^{r_2 t}.$$

$$Y_p = A_2^{-1} B$$

$$A^{-1} = \frac{\text{Adj } A_2}{|A|}$$

$$Y_T = Y_c + Y_p.$$

Simultaneous Differential & Difference Equations:-

Solving Simultaneous Dynamic Equations:-

The methods of solving simultaneous differential & simultaneous difference equations are quite similar.

The following system of linear difference equations

$$x_{t+1} + 6x_t + 9y_t = 4$$
$$y_{t+1} - x_t = 0$$

is known as simultaneous Difference Equation.

Dynamic Input-Output Models:-

The context was static and the ~~and~~ it was to solve a simultaneous-equation system for the equilibrium output levels of all industries. When certain additional economic considerations are incorporated into the model, the input-output system can take on a dynamic character, and there will then result a difference or differential equation system.

Inflation-Unemployment Models:-

Simultaneous differential equations :- The inflation-unemployment model was presented in the continuous-time framework via the following three equations:

$$p = \alpha - T - \beta U + g\pi \quad (\alpha, \beta > 0; 0 < g \leq 1).$$

$$\frac{d\pi}{dt} = j(p - \pi) \quad (0 < j \leq 1).$$

$$\frac{dU}{dt} = -k(\mu - p) \quad (k > 0).$$

Simultaneous Difference equations:- The simultaneous-equation treatment of the inflation-unemployment model in discrete time is similar in spirit to the preceding continuous-time discussion, consists of three equations:

$$p_t = \alpha - T - \beta U_t + g\,\pi_t.$$

$$\pi_{t+1} - \pi_t = j(p_t - \pi_t)$$

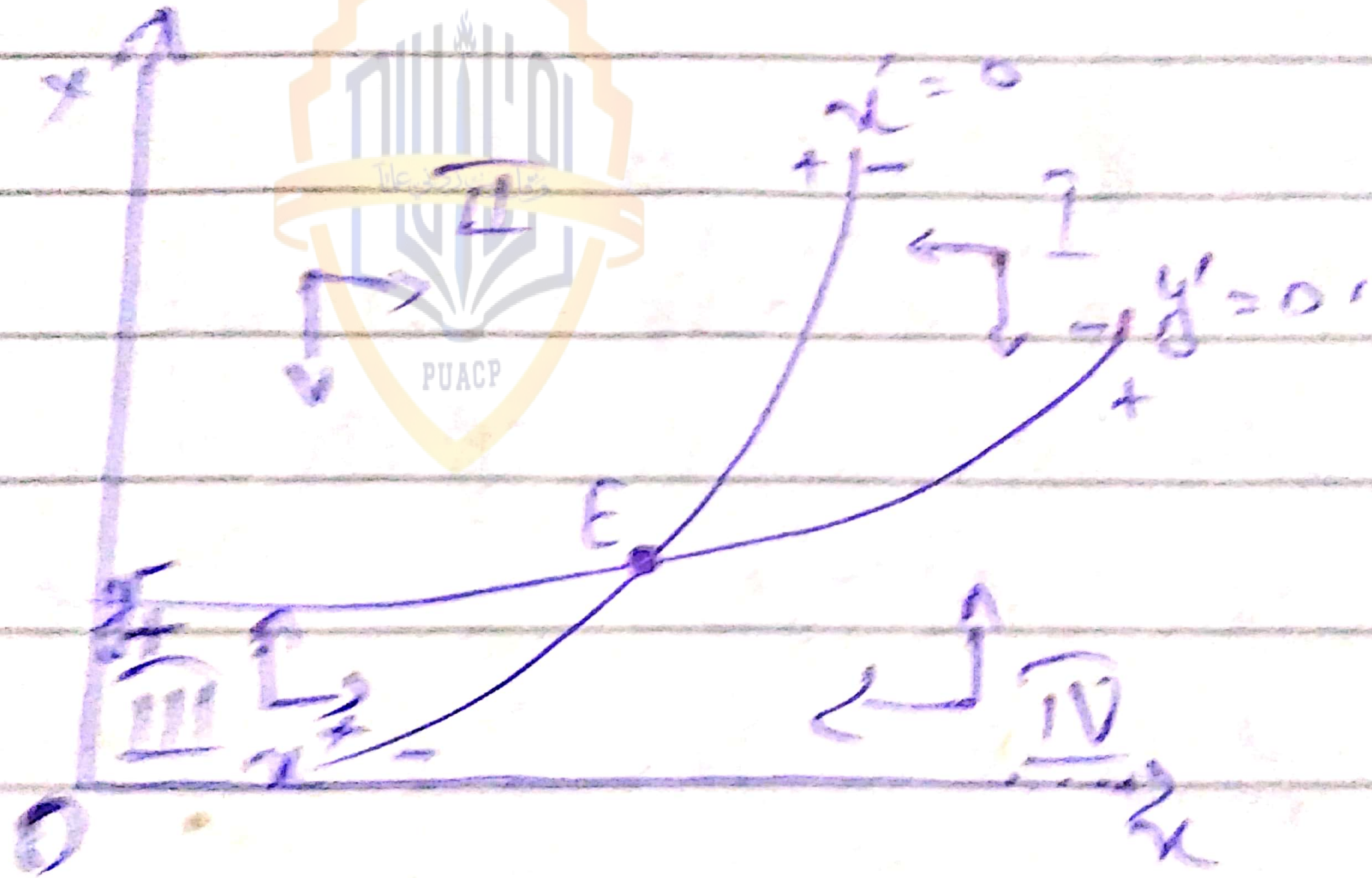
$$U_{t+1} - U_t = -k(\mu - p_{t+1}).$$

Two Variables Phase Diagrams:-

In two variables phase diagrams we discuss the qualitative-graphic analysis of a nonlinear differential-equation system as well as the first-order differential-equation system in two variables.

$$x'(t) = f(x, y).$$

$$y'(t) = g(x, y).$$



x
x' = 0
y' = 0
I
II
III
IV
E
O
x